**VENKATA SURENDRA REDDY G**
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## PROFESSIONAL SUMMARY

Senior AI/ML Engineer with over five years of experience specialized in GenAI and LLM(OpenAI, Hugging Face) technologies, enhancing customer experience stacks. Possessed strong proficiency in SQL database operations, Python, and JavaScript, building LLM-based applications using APIs, embeddings, and vector databases. Successfully designed and implemented technical solutions, translating complex business needs into robust architectures that enhanced customer experience stacks. Leveraged Cloud services, CDPs, CRM, and E-commerce platforms, proposing architectures using agent-based orchestration to collect, activate, and use customer data. Adopted AI-driven practices, building hands-on LLM-based prototypes and communicating technical decisions effectively to stakeholders for impactful outcomes.

## TECHNICAL SKILLS

**Programming Languages & Frameworks:** Python, SQL, JavaScript, TypeScript, Java, FastAPI, REST APIs, Microservices Architecture
**Generative AI & LLM Concepts:** Generative AI, Large Language Models (LLMs) (OpenAI, Hugging Face), Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Vector Search, Context Construction, Structured Output Generation
**Machine Learning:** Machine Learning, Deep Learning, LangChain, Agent-Oriented Workflows, Tool Integration, Time Series Forecasting, Feature Engineering, Explainable AI (SHAP), Model Evaluation
**Databases & Data Storage:** Vector Databases, PostgreSQL, SQL Server, Snowflake, MongoDB, Parquet
**Cloud Platforms:** AWS(S3, Lambda), Azure
**Data Engineering:** ETL/ELT Pipelines, Data Modeling, PySpark, Large-Scale Data Processing, Data Validation, Data Quality
**DevOps & MLOps:** MLOps, CI/CD Pipelines, Docker, GitHub, GitLab, Jenkins, MLflow, Model Versioning, Model Monitoring
**Business Platforms & Analytics:** CDPs, CRM, E-commerce Platforms, Power BI, Tableau, Data Storytelling

## WORK EXPERIENCE

**CHiEAC | AI & Data Science Fellow | Jul 2025 - Present**

- Designed and deployed enterprise-grade AI solutions for financial risk analytics, integrating machine learning models with LLMs (OpenAI/Hugging Face) and RAG pipelines to deliver explainable, audit-ready insights
- Developed LLM-powered workflows using LangChain for prompt orchestration, context injection, and structured output generation to support compliance and business decision-making
- Built and fine-tuned deep learning models using PyTorch/TensorFlow for risk prediction and integrated outputs into downstream LLM reasoning pipelines
- Implemented FastAPI-based microservices and REST APIs to integrate AI solutions with internal applications, data platforms, and client-facing systems
- Designed retrieval pipelines leveraging vector search and ElasticSearch to enable high-performance contextual retrieval for LLM-based reasoning
- Integrated Kafka-based streaming pipelines to process real-time financial data and support low-latency inference workflows
- Ensured security, auditability, and regulatory compliance by incorporating SHAP-based explainability, structured outputs, and governed data pipelines
- Orchestrated CI/CD pipelines and MLOps workflows on AWS (S3, Lambda, Airflow) for automated model training, validation, and deployment
- Participated in Agile ceremonies, collaborated with cross-functional teams for requirement gathering, design, and delivery, and supported UAT, defect triage, and production deployment

**DePaul University | Machine Learning Consultant | Mar 2025 – Jun 2025**

- Built LabelLanche GPT, a GenAI-powered RAG system using LangChain and OpenAI APIs to automate classification and enrichment of 200K+ CRM records

- Developed LLM-based NLP pipelines for structured data extraction, classification, and automation of enterprise workflows
- Implemented vector databases and embeddings for semantic retrieval, improving accuracy and handling complex, ambiguous inputs
- Designed API-based integrations to connect AI pipelines with CRM systems and Snowflake data platforms
- Containerized and deployed the system using Docker and CI/CD pipelines, ensuring scalability and production readiness
- Optimized data pipelines and ETL workflows for large-scale data processing and integration with enterprise data platforms
- Ensured data security and governance using row-level security (RLS) and structured access controls
- Collaborated in Agile environments and supported testing, validation, and UAT cycles for enterprise deployment

**Quari Ice | Machine Learning Engineer | Jun 2024 – Mar 2025**

- Developed and deployed AI/ML models using TensorFlow and PyTorch for demand forecasting and supply chain optimization across 150+ retail locations
- Built data pipelines and microservices-based architectures to integrate AI solutions with business systems and operational workflows
- Implemented REST APIs and backend services to expose model predictions for downstream applications and decision systems
- Designed model evaluation and monitoring frameworks to ensure performance, reliability, and scalability in production environments
- Automated CI/CD pipelines using Jenkins and integrated MLflow for model versioning, monitoring, and lifecycle management
- Leveraged cloud platforms (AWS) for scalable deployment and data storage, ensuring high availability and performance
- Collaborated with business stakeholders to translate requirements into AI-driven solutions and supported UAT, defect resolution, and production rollout

**WNS Global Solutions | Data Scientist | May 2021 – Jun 2023**

- Designed and implemented a scalable data and analytics platform using Python and SQL, integrating sensor telemetry and ERP production data to support downstream AI/ML use cases in manufacturing operations.
- Built robust ETL pipelines using SQL and Python to process high-volume operational data, enabling reliable data availability for predictive modeling and analytics workflows.
- Developed a predictive maintenance system using Python and regression-based ML techniques, identifying early indicators of equipment failure and reducing unexpected production downtime by 12%.
- Engineered a centralized data repository using MongoDB to manage unstructured technical documentation and operational insights, supporting knowledge accessibility across teams.
- Automated recurring reporting and analytics workflows using Python, reducing manual data preparation effort by 40+ hours per month.
- Designed real-time operational dashboards using Power BI to monitor production KPIs and shift-level performance, enabling data-driven decision-making on the factory floor.
- Proposed scalable data solutions, collaborating with cross-functional teams to translate manufacturing data challenges into scalable data solutions, supporting broader GenAI/ML initiatives.
- Maintained version-controlled development workflows using GitHub, ensuring reproducibility and structured deployment of analytics solutions.

## EDUCATION

Master of Science, Business Analytics, DePaul University (Jun 2025)